ICE4405P Digital Signal Processing
Lab 1 Report

Convolution, DTFT, Z-Transform

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Contents

1 Problem 1: Simulating a Recording Studio's Reverberation

1.1 Given

$$h[n] = \delta[n] + 0.5\delta[n-1] + 0.3\delta[n-2]$$

$$x[n] = [2,\ 4,\ 6,\ 4,\ 2,\ 0]$$

1.2 Method

Compute the reverberated output using discrete-time convolution:

$$y[n] = x[n] * h[n]$$

MATLAB command:

```
y = conv(x, h);
```

1.3 Result

$$y[n] = [2,\ 5,\ 8.6,\ 8.2,\ 5.8,\ 2.2,\ 0.6,\ 0]$$

1.4 Figure

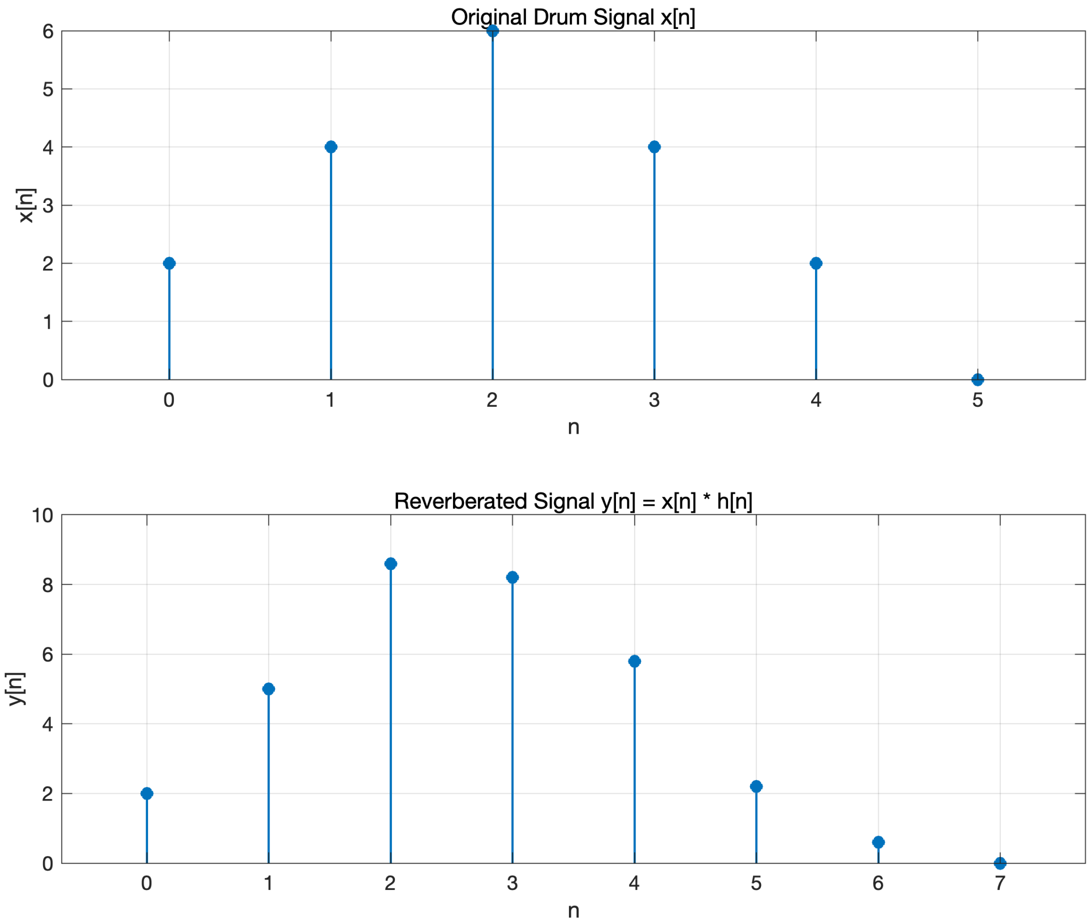

Figure 1: Problem 1: Original and reverberated signals

1.5 Discussion

The output sequence is longer than the original sequence and shows smoother decay, consistent with reverberation introduced by delayed and attenuated reflections.

2 Problem 2: Simulating Echo in an Open Space

2.1 Given

$$x[n] = 0.8^n, \quad 0 \leq n \leq 10$$

$$h[n] = \delta[n] + 0.5\delta[n-1] + 0.3\delta[n-2]$$

2.2 Method

Compute echoed signal:

$$y[n] = x[n] * h[n]$$

MATLAB command:

```
n = 0:10;
x = 0.8.^n;
y = conv(x, h);
```

2.3 Result

First several samples:

$$y[n] = [1,\ 1.3,\ 1.34,\ 1.072,\ 0.8576,\ \cdots]$$

2.4 Figure

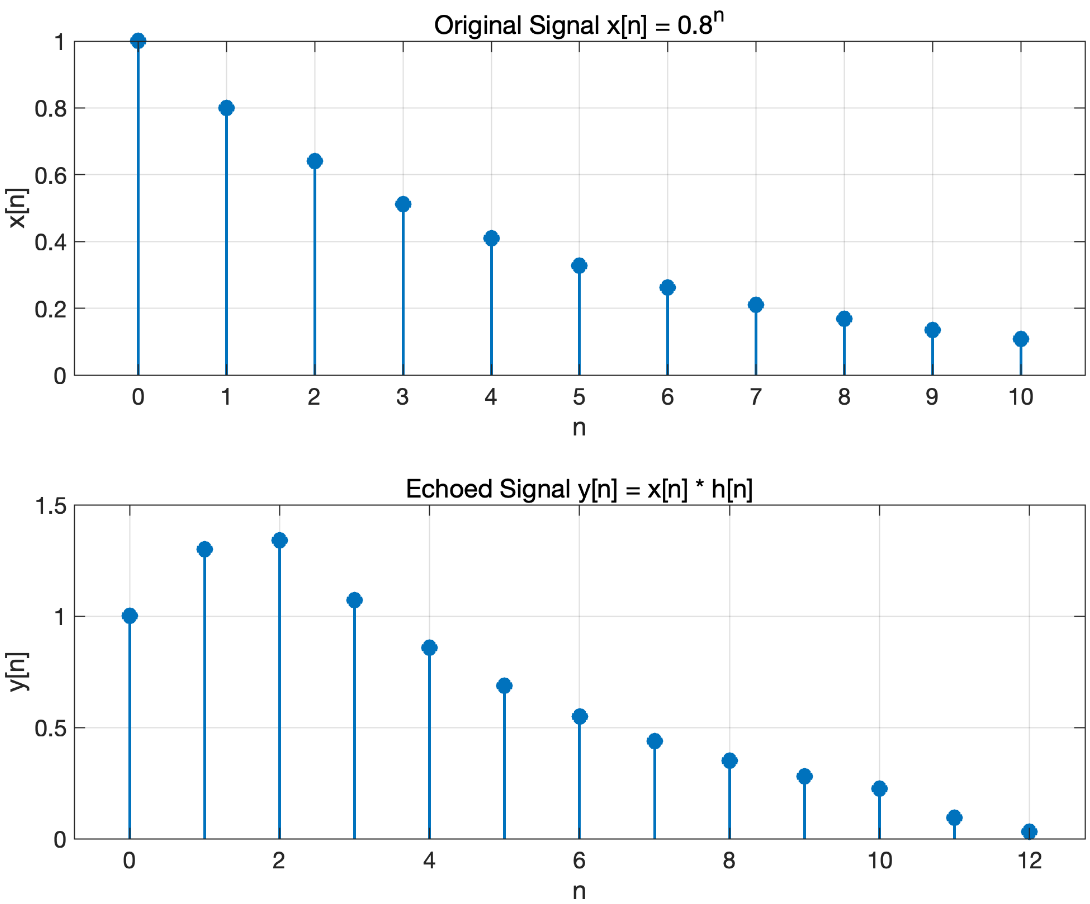

Figure 2: Problem 2: Original and echoed signals

2.5 Discussion

The echoed signal has larger early samples and a longer tail than the original decaying sequence, matching expected multi-path reflection behavior.

3 Problem 3: Spectrum Analysis of a Speech Signal (DTFT)

3.1 Given

$$x[n] = 0.8^n, \quad 0 \le n \le 10$$

3.2 DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

For this finite-length sequence:

$$X(\omega) = \sum_{n=0}^{10} (0.8)^n e^{-j\omega n}$$

3.3 Method

Numerically evaluate $X(\omega)$ over $\omega \in [-\pi, \pi]$, then plot:

$$|X(\omega)|, \quad \angle X(\omega)$$

3.4 Figure

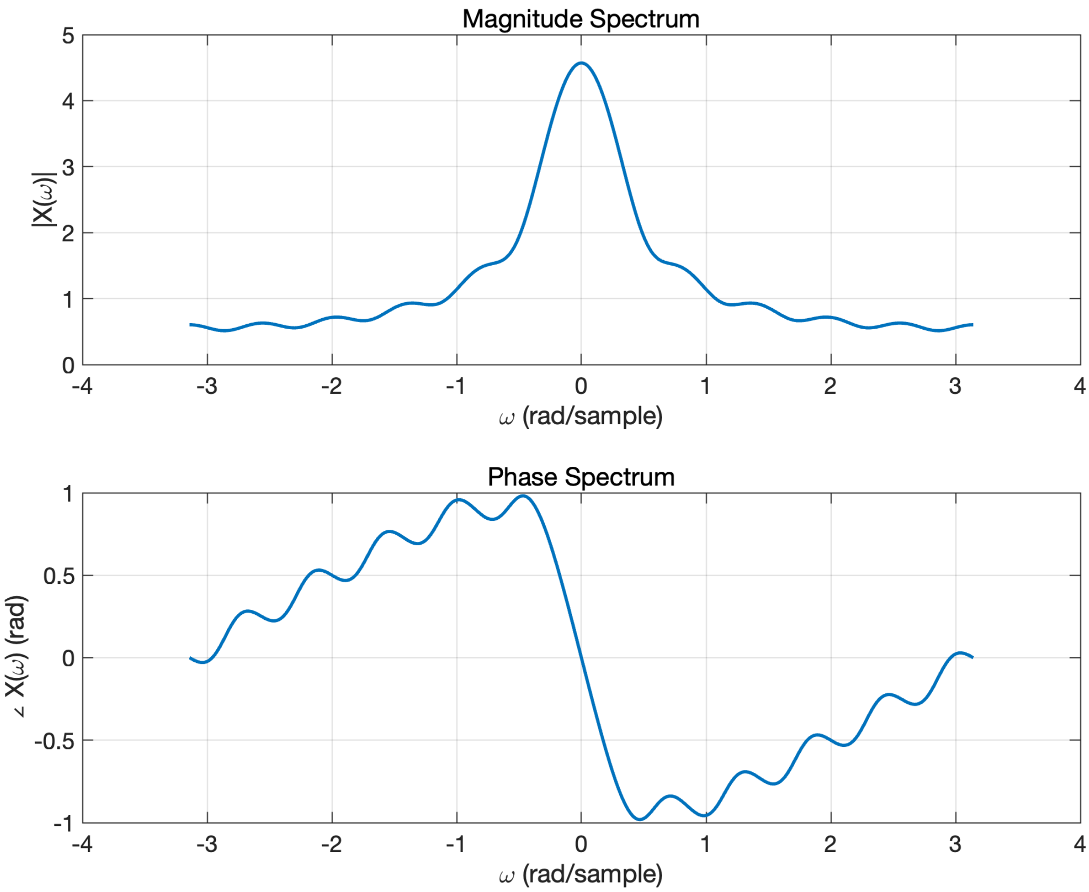

Figure 3: Problem 3: Magnitude and phase spectra of DTFT

3.5 Discussion

The magnitude is strongest near $\omega = 0$, indicating that low-frequency components dominate this exponentially decaying sequence.

4 Problem 4: High-Pass Filter for Speech Enhancement

4.1 Given

$$h[n] = (-0.5)^n u[n]$$

4.2 Z-Transform and ROC

$$H(z) = \sum_{n=0}^{\infty} (-0.5)^n z^{-n} = \frac{1}{1 + 0.5z^{-1}}$$

Pole location:

$$z = -0.5$$

Region of convergence:

$$\text{ROC} : |z| > 0.5$$

4.3 Stability and Interpretation

Since $|-0.5| < 1$ and the ROC includes the unit circle, the filter is stable. Also, from the frequency response:

$$|H(e^{j\pi})| > |H(e^{j0})|$$

it emphasizes higher frequencies, which is consistent with high-pass behavior and useful for suppressing low-frequency interference in speech enhancement.

4.4 Figure

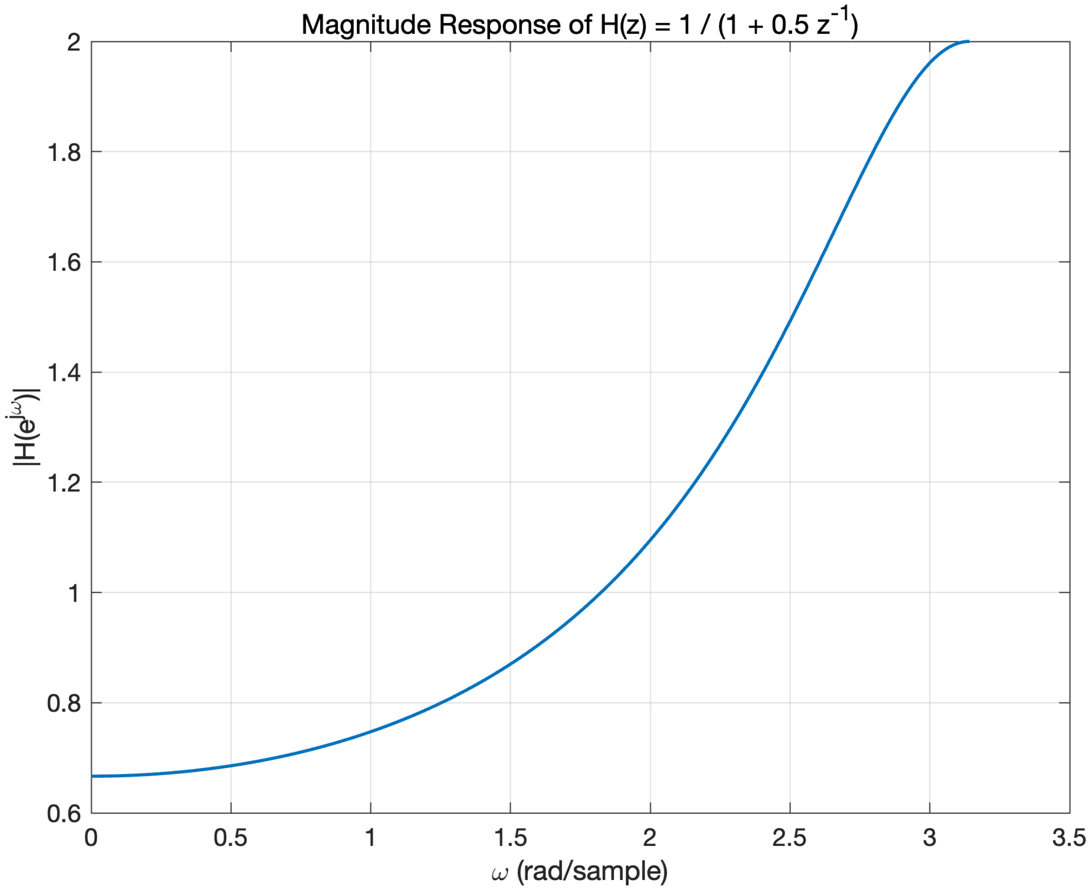

Figure 4: Problem 4: Magnitude response of high-pass filter

5 Problem 5: Stability Analysis of a Low-Pass Filter

5.1 Given

$$h[n] = (0.7)^n u[n]$$

5.2 Z-Transform and ROC

$$H(z) = \sum_{n=0}^{\infty} (0.7)^n z^{-n} = \frac{1}{1 - 0.7z^{-1}}$$

Pole location:

$$z = 0.7$$

Region of convergence:

$$\text{ROC} : |z| > 0.7$$

5.3 Stability and Comparison

Since $|0.7| < 1$ and the unit circle is included in ROC, this filter is stable. For this filter:

$$|H(e^{j0})| > |H(e^{j\pi})|$$

so it has low-pass behavior. Compared with Problem 4, Problem 5 is better for smoothing and low-frequency preservation, while Problem 4 is better for suppressing low-frequency components.

5.4 Figure

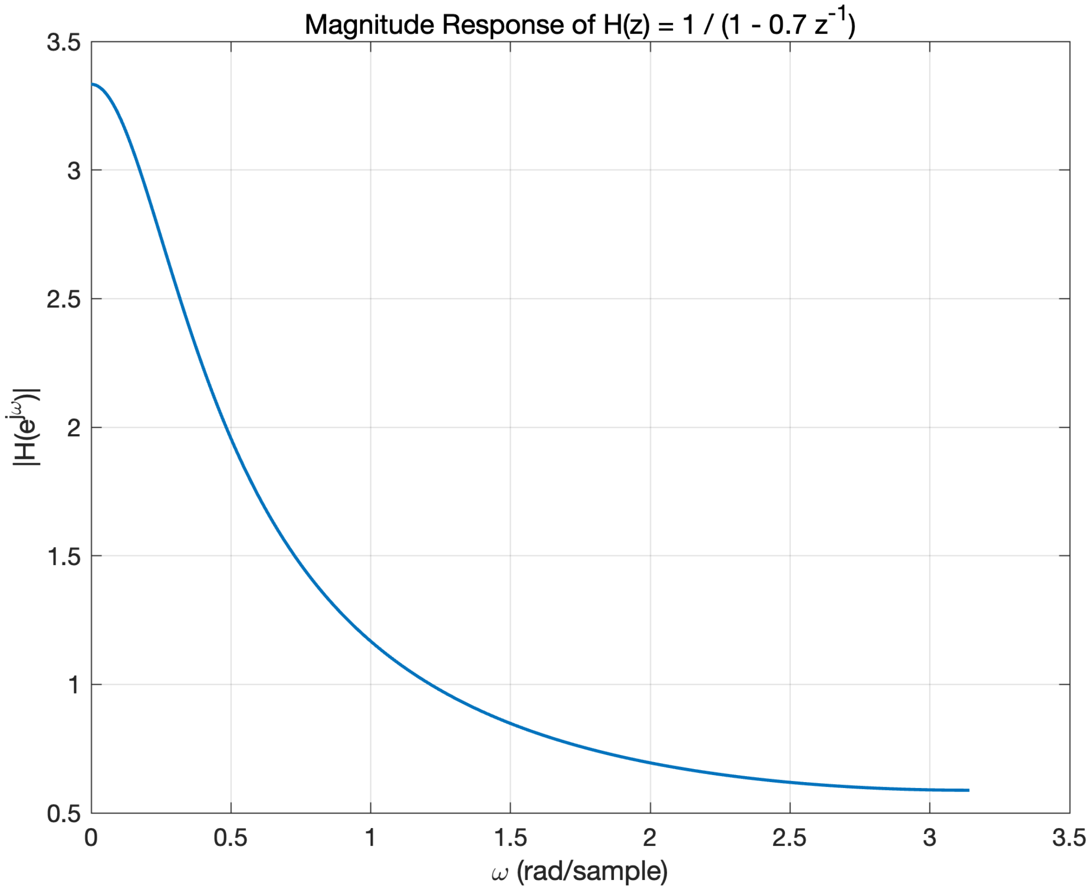

Figure 5: Problem 5: Magnitude response of low-pass filter

6 Conclusion

This lab verifies three core DSP tools:

1. Convolution for time-domain system effects (reverberation/echo),
2. DTFT for frequency-domain analysis,

3. Z-transform for pole/ROC-based stability and filter interpretation.

Both filters in this lab are stable but provide different frequency-selective behavior for speech processing tasks.